

Dynamic allocation + scheduling vs best measured allocation

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Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Balanced compute	2	(1, 1)	(1, 1)	2.5314	2.4157	+0.1157	+4.8%
Balanced compute	4	(2, 1)	(3, 3)	2.4838	1.8355	+0.6484	+35.3%
Balanced compute	6	(2, 1)	(5, 5)	2.5641	1.2662	+1.2979	+102.5%
Balanced compute	8	(3, 3)	(5, 6)	1.9651	1.1565	+0.8086	+69.9%
Chunk edge	2	(1, 1)	(1, 1)	0.0120	0.0112	+0.0009	+7.6%
Chunk edge	4	(1, 1)	(1, 1)	0.0121	0.0112	+0.0009	+8.0%
Chunk edge	6	(1, 1)	(1, 1)	0.0120	0.0111	+0.0009	+7.8%
Chunk edge	8	(1, 1)	(1, 1)	0.0120	0.0112	+0.0008	+7.3%
Large arrays	2	(1, 1)	(1, 1)	5.7699	5.5603	+0.2096	+3.8%
Large arrays	4	(2, 1)	(3, 3)	5.2452	5.2017	+0.0434	+0.8%
Large arrays	6	(2, 1)	(5, 3)	5.5470	5.0884	+0.4586	+9.0%
Large arrays	8	(3, 3)	(5, 3)	5.3058	5.1203	+0.1855	+3.6%
Compute loops + short tasks	4	(2, 1) *	(3, 3)	1.5604	0.9356	+0.6247	+66.8%
Compute loops + short tasks	6	(1, 1)	(5, 5)	1.4505	0.6713	+0.7792	+116.1%

Gap = (model time / best measured time - 1) x 100. Times are medians of process medians.

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Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Mixed: long ordinary	4	(1, 1)	(1, 1)	0.6817	0.6712	+0.0104	+1.6%
Unequal loops + short tasks	4	(2, 1)	(3, 1)	0.3823	0.2826	+0.0997	+35.3%
Unequal loops + short tasks	6	(1, 1)	(5, 1)	0.5408	0.2245	+0.3163	+140.9%
Mixed: medium ordinary	4	(1, 1)	(3, 3)	0.2626	0.2480	+0.0147	+5.9%
Mixed: medium ordinary	6	(1, 1)	(5, 5)	0.2896	0.2588	+0.0307	+11.9%
One long ordinary task	4	(1, 1)	(1, 1)	0.4446	0.4160	+0.0286	+6.9%
Mixed: one ordinary	2	(1, 1)	(1, 1)	0.2266	0.2209	+0.0057	+2.6%
Mixed: one ordinary	4	(1, 1)	(3, 1)	0.2236	0.2118	+0.0118	+5.6%
Mixed: one ordinary	6	(1, 1)	(5, 1)	0.2273	0.2148	+0.0125	+5.8%
Three short ordinary tasks	2	(1, 1)	(1, 1)	0.2419	0.2400	+0.0019	+0.8%
Three short ordinary tasks	4	(1, 1)	(3, 1)	0.2413	0.2141	+0.0271	+12.7%
Three short ordinary tasks	6	(1, 1)	(5, 1)	0.2236	0.2110	+0.0126	+6.0%
Mixed: two ordinary	2	(1, 1)	(1, 1)	0.2451	0.2430	+0.0021	+0.9%
Mixed: two ordinary	4	(1, 1)	(3, 1)	0.2441	0.2323	+0.0118	+5.1%

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Mixed: two ordinary	6	(1, 1)	(5, 1)	0.2301	0.2116	+0.0185	+8.8%
Short	2	(1, 1)	(1, 1)	0.0489	0.0388	+0.0101	+26.0%
Short	4	(1, 1)	(1, 1)	0.0414	0.0388	+0.0026	+6.6%
Short	6	(1, 1)	(1, 1)	0.0413	0.0388	+0.0026	+6.6%
Short	8	(1, 1)	(1, 1)	0.0414	0.0390	+0.0024	+6.2%
Tiny	2	(1, 1)	(1, 1)	0.0024	0.0016	+0.0008	+51.5%
Tiny	4	(1, 1)	(1, 1)	0.0024	0.0016	+0.0008	+51.4%
Tiny	6	(1, 1)	(1, 1)	0.0024	0.0016	+0.0008	+51.5%
Tiny	8	(1, 1)	(1, 1)	0.0024	0.0016	+0.0008	+50.8%
Trip skew	2	(1, 1)	(1, 1)	4.1272	3.9993	+0.1278	+3.2%
Trip skew	4	(2, 1)	(3, 3)	2.6730	1.7232	+0.9498	+55.1%
Trip skew	6	(4, 1)	(5, 5)	1.6835	1.1640	+0.5195	+44.6%
Trip skew	8	(6, 1)	(6, 5)	1.6652	1.0428	+0.6224	+59.7%
Uneven chunks	2	(1, 1)	(1, 1)	0.0754	0.0736	+0.0018	+2.5%

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Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Uneven chunks	4	(1, 1)	(1, 1)	0.0752	0.0749	+0.0003	+0.4%
Uneven chunks	6	(1, 1)	(1, 1)	0.0754	0.0754	+0.0000	+0.0%
Uneven chunks	8	(1, 1)	(1, 1)	0.0759	0.0750	+0.0010	+1.3%
Work skew	2	(1, 1)	(1, 1)	5.8529	5.7599	+0.0930	+1.6%
Work skew	4	(2, 1)	(3, 3)	3.3130	2.2773	+1.0358	+45.5%
Work skew	6	(4, 1)	(5, 5)	1.8298	1.5258	+0.3040	+19.9%
Work skew	8	(6, 1)	(6, 4)	1.4606	1.3805	+0.0800	+5.8%
Work skew reversed	2	(1, 1)	(1, 1)	6.0509	5.9509	+0.1001	+1.7%
Work skew reversed	4	(1, 2)	(3, 3)	3.4604	2.4535	+1.0069	+41.0%
Work skew reversed	6	(1, 4)	(5, 5)	2.0148	1.4709	+0.5439	+37.0%
Work skew reversed	8	(1, 6)	(3, 7)	1.4778	1.3948	+0.0830	+6.0%

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Ordinary-first reference vs best measured allocation

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Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Balanced compute	2	(1, 1)	(1, 1)	2.4806	2.4157	+0.0648	+2.7%
Balanced compute	4	(2, 1)	(3, 3)	2.5051	1.8355	+0.6696	+36.5%
Balanced compute	6	(2, 1) *	(5, 5)	2.6918	1.2662	+1.4256	+112.6%
Balanced compute	8	(3, 3)	(5, 6)	2.1356	1.1565	+0.9791	+84.7%
Chunk edge	2	(1, 1)	(1, 1)	0.1540	0.0112	+0.1428	+1278.8%
Chunk edge	4	(1, 1)	(1, 1)	0.1793	0.0112	+0.1681	+1505.3%
Chunk edge	6	(1, 1)	(1, 1)	0.1771	0.0111	+0.1660	+1491.0%
Chunk edge	8	(1, 1)	(1, 1)	0.1730	0.0112	+0.1618	+1443.1%
Large arrays	2	(1, 1)	(1, 1)	5.5992	5.5603	+0.0389	+0.7%
Large arrays	4	(2, 1)	(3, 3)	5.2475	5.2017	+0.0458	+0.9%
Large arrays	6	(2, 1)	(5, 3)	6.1471	5.0884	+1.0587	+20.8%
Large arrays	8	(3, 3)	(5, 3)	5.5737	5.1203	+0.4534	+8.9%
Compute loops + short tasks	4	(1, 2) *	(3, 3)	1.5698	0.9356	+0.6342	+67.8%
Compute loops + short tasks	6	(1, 1)	(5, 5)	1.3765	0.6713	+0.7052	+105.0%

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Mixed: long ordinary	4	(1, 1)	(1, 1)	0.7844	0.6712	+0.1132	+16.9%
Unequal loops + short tasks	4	(2, 1)	(3, 1)	0.7055	0.2826	+0.4229	+149.6%
Unequal loops + short tasks	6	(1, 1)	(5, 1)	0.7499	0.2245	+0.5254	+234.0%
Mixed: medium ordinary	4	(1, 1)	(3, 3)	0.4983	0.2480	+0.2503	+100.9%
Mixed: medium ordinary	6	(1, 1)	(5, 5)	0.3185	0.2588	+0.0597	+23.1%
One long ordinary task	4	(1, 1)	(1, 1)	0.7532	0.4160	+0.3373	+81.1%
Mixed: one ordinary	2	(1, 1)	(1, 1)	0.2560	0.2209	+0.0350	+15.9%
Mixed: one ordinary	4	(1, 1)	(3, 1)	0.3373	0.2118	+0.1255	+59.3%
Mixed: one ordinary	6	(1, 1)	(5, 1)	0.2886	0.2148	+0.0738	+34.3%
Three short ordinary tasks	2	(1, 1)	(1, 1)	0.4360	0.2400	+0.1960	+81.7%
Three short ordinary tasks	4	(1, 1)	(3, 1)	0.4555	0.2141	+0.2413	+112.7%
Three short ordinary tasks	6	(1, 1)	(5, 1)	0.2869	0.2110	+0.0758	+35.9%
Mixed: two ordinary	2	(1, 1)	(1, 1)	0.4336	0.2430	+0.1906	+78.5%
Mixed: two ordinary	4	(1, 1)	(3, 1)	0.3160	0.2323	+0.0837	+36.0%

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Mixed: two ordinary	6	(1, 1)	(5, 1)	0.3199	0.2116	+0.1083	+51.2%
Short	2	(1, 1)	(1, 1)	0.1682	0.0388	+0.1294	+333.7%
Short	4	(1, 1)	(1, 1)	0.1861	0.0388	+0.1473	+379.8%
Short	6	(1, 1)	(1, 1)	0.1895	0.0388	+0.1508	+388.9%
Short	8	(1, 1)	(1, 1)	0.2022	0.0390	+0.1632	+418.2%
Tiny	2	(1, 1)	(1, 1)	0.1431	0.0016	+0.1416	+9016.5%
Tiny	4	(1, 1)	(1, 1)	0.1682	0.0016	+0.1666	+10606.0%
Tiny	6	(1, 1)	(1, 1)	0.1656	0.0016	+0.1640	+10447.8%
Tiny	8	(1, 1)	(1, 1)	0.1841	0.0016	+0.1825	+11486.9%
Trip skew	2	(1, 1)	(1, 1)	4.1302	3.9993	+0.1309	+3.3%
Trip skew	4	(2, 1)	(3, 3)	2.6350	1.7232	+0.9118	+52.9%
Trip skew	6	(4, 1)	(5, 5)	2.5085	1.1640	+1.3445	+115.5%
Trip skew	8	(6, 1)	(6, 5)	1.7217	1.0428	+0.6789	+65.1%
Uneven chunks	2	(1, 1)	(1, 1)	0.2193	0.0736	+0.1458	+198.1%

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Uneven chunks	4	(1, 1)	(1, 1)	0.2392	0.0749	+0.1642	+219.1%
Uneven chunks	6	(1, 1)	(1, 1)	0.2457	0.0754	+0.1703	+225.7%
Uneven chunks	8	(1, 1)	(1, 1)	0.2479	0.0750	+0.1729	+230.7%
Work skew	2	(1, 1)	(1, 1)	5.7811	5.7599	+0.0213	+0.4%
Work skew	4	(2, 1)	(3, 3)	3.2292	2.2773	+0.9520	+41.8%
Work skew	6	(4, 1)	(5, 5)	3.3908	1.5258	+1.8649	+122.2%
Work skew	8	(6, 1)	(6, 4)	2.9037	1.3805	+1.5231	+110.3%
Work skew reversed	2	(1, 1)	(1, 1)	5.9473	5.9509	-0.0035	-0.1%
Work skew reversed	4	(1, 2)	(3, 3)	3.5445	2.4535	+1.0910	+44.5%
Work skew reversed	6	(1, 4)	(5, 5)	1.9668	1.4709	+0.4960	+33.7%
Work skew reversed	8	(1, 6)	(3, 7)	2.7712	1.3948	+1.3764	+98.7%

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